COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	2026-10-29	**DATE SUBMITTED:**	2026-10-29
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ACTIVITY NO. **Final Hands-on Skills Exam**

1. PROCEDURE OUTPUT

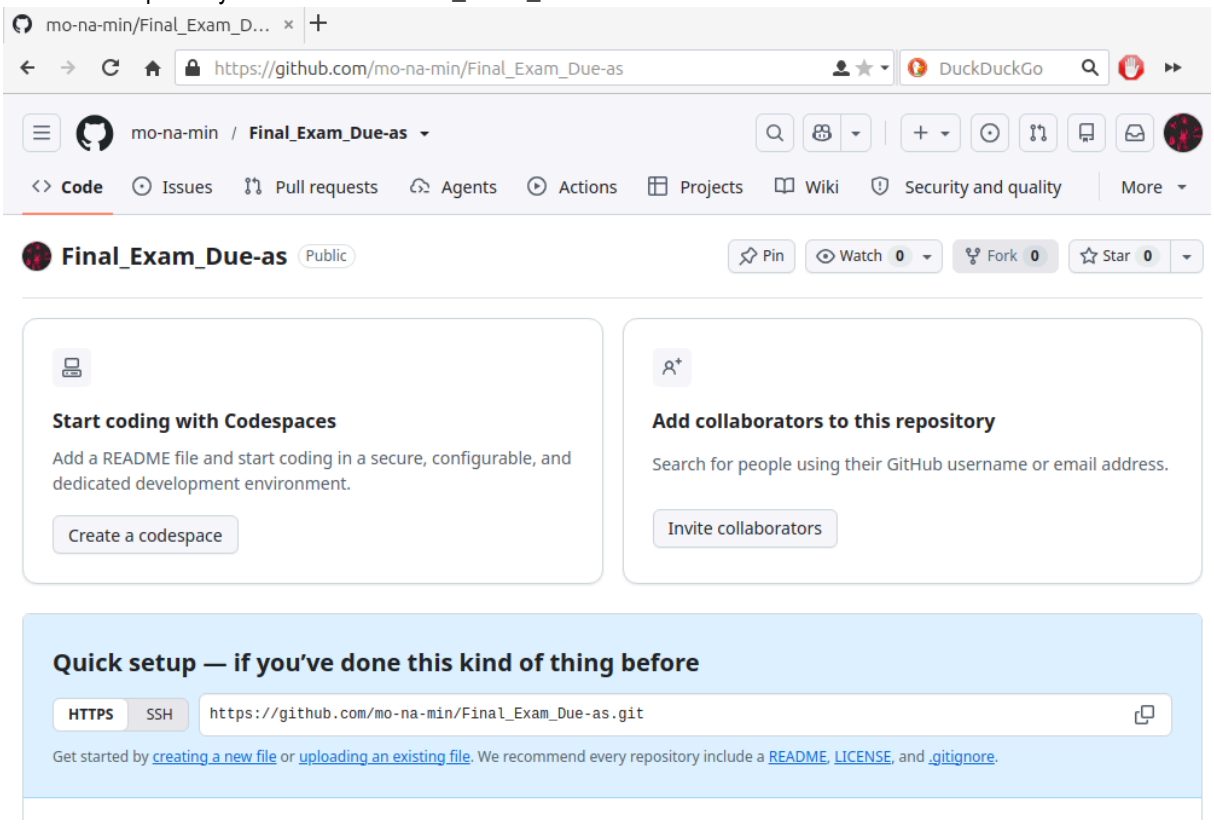
Tools Needed:

1. VM with Ubuntu, CentOS and Ansible installed

2. Web browser

Procedure:

1. Create a repository and label it as "Final_Exam_Surname"



2. Clone your new repository in your VM

```
localserver@workstation:~$ git clone https://github.com/mo-na-min/Final_Exam_Due-as.git
Cloning into 'Final_Exam_Due-as'...
warning: You appear to have cloned an empty repository.
```

3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.

tree

```
localserver@workstation:~/Final_Exam_Due-as$ tree
.
├── config.yaml
├── inventory.ini
├── roles
│   ├── elasticsearch
│   │   └── tasks
│   │       └── main.yml
│   ├── enterprise_service
│   │   └── tasks
│   │       └── main.yml
│   ├── kibana
│   │   └── tasks
│   │       └── main.yml
│   └── logstash
│       └── tasks
│           └── main.yml
└── site.yml

10 directories, 7 files
```

config.yaml

```
---
#enterprise service
enterprise_service_package: nginx
enterprise_service_port: 80

#monitoring stack
elasticsearch_port: 9200
kibana_port: 5601
```

inventory.ini

```
[all_servers]
192.168.56.104 ansible_user=ubuntu
192.168.56.105 ansible_user=ubuntu
192.168.56.108 ansible_user=centos
```

site.yml

```
---
- name: Deploy Enterprise Service and Elastic Monitoring Stack Across All Servers
  hosts: all_servers
  become: true
  vars_files:
    - config.yaml

  roles:
    - role: enterprise_service
    - role: elasticsearch
    - role: logstash
    - role: kibana
```

3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers

Enterprise Service

```
---
- name: Install Enterprise Service (Debian/Ubuntu)
  ansible.builtin.apt:
    name: "{{ enterprise_service_package }}"
    state: present
    update_cache: yes
  when: ansible_facts['os_family'] == "Debian"

- name: Install Enterprise Service (RedHat/CentOS)
  ansible.builtin.yum:
    name: "{{ enterprise_service_package }}"
    state: present
  when: ansible_facts['os_family'] == "RedHat"

- name: Ensure Enterprise Service is enabled and running
  ansible.builtin.systemd:
    name: "{{ enterprise_service_package }}"
    state: started
    enabled: yes
```

3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)

Monitoring Tool

Debian

```
---
# Debian
- name: Clean APT cache
  ansible.builtin.apt:
    clean: yes
  when: ansible_facts['os_family'] == "Debian"

- name: Install prerequisites (Debian/Ubuntu)
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - openjdk-11-jdk
      - gpg
      - curl
    state: present
    update_cache: yes
    lock_timeout: 180
  environment:
    DEBIAN_FRONTEND: noninteractive
  when: ansible_facts['os_family'] == "Debian"

- name: Create Keyrings directory
  ansible.builtin.file:
    path: /etc/apt/keyrings
    state: directory
    mode: '0755'
  when: ansible_facts['os_family'] == "Debian"
```

```
- name: Add Elastic GPG key (Debian/Ubuntu)
  ansible.builtin.get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /etc/apt/keyrings/elasticsearch.asc
    mode: '0644'
  when: ansible_facts['os_family'] == "Debian"

- name: Add Elastic APT repository (Debian/Ubuntu)
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/etc/apt/keyrings/elasticsearch.asc] https://artifacts.elastic.co/packages/8.x/apt stable main"
    state: present
    filename: elastic-8.x
  when: ansible_facts['os_family'] == "Debian"

- name: Install Elasticsearch (Debian/Ubuntu)
  ansible.builtin.apt:
    name: elasticsearch
    state: present
    update_cache: yes
    lock_timeout: 180
  environment:
    DEBIAN_FRONTEND: noninteractive
  when: ansible_facts['os_family'] == "Debian"
```

CentOS

```
# CentOS
- name: Import Elastic GPG Key (RedHat/CentOS)
  ansible.builtin.rpm_key:
    state: present
    key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
  when: ansible_facts['os_family'] == "RedHat"

- name: Add Elastic YUM Repository (RedHat/CentOS)
  ansible.builtin.yum_repository:
    name: elastic-8.x
    description: Elastic repository for 8.x packages
    baseurl: https://artifacts.elastic.co/packages/8.x/yum
    gpgcheck: yes
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: yes
  when: ansible_facts['os_family'] == "RedHat"

- name: Install Elasticsearch (RedHat/CentOS)
  ansible.builtin.yum:
    name: elasticsearch
    state: present
  when: ansible_facts['os_family'] == "RedHat"
```

Configuration

```
# Configuration
- name: Configure Elasticsearch network binding
  ansible.builtin.lineinfile:
    path: /etc/elasticsearch/elasticsearch.yml
    regexp: '^#?network.host:'
    line: 'network.host: 0.0.0.0'

- name: Start and enable Elasticsearch service
  ansible.builtin.systemd:
    name: elasticsearch
    state: started
    enabled: yes
```

4.4 Change Motd as "Ansible Managed by <username>"

site.yaml

```
- name: MOTD all servers
  hosts: all_servers
  become: true
  tasks:
    - name: set MOTD
      ansible.builtin.copy:
        content: "Ansible Managed by {{ ansible_user_id }}\n"
        dest: /etc/motd/
        owner: root
        group: root
        mode: '0644'
```

4. Push and commit your files in GitHub

GitHub

5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)

ansible-playbook

```
localserver@workstation:~/Final_Exam_Due-as$ ansible-playbook -i inventory.ini site.yml --extra-vars "@config.yaml" --ask-becom
e-pass
BECOME password:

PLAY [Deploy Enterprise Service and Elastic Monitoring Stack Across All Servers] ***********************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.108]
ok: [192.168.56.105]
ok: [192.168.56.104]

TASK [enterprise_service : Install NGINX (Debian/Ubuntu)] **********************************************************
skipping: [192.168.56.108]
ok: [192.168.56.105]
ok: [192.168.56.104]

TASK [enterprise_service : Install NGINX (RedHat/CentOS)] **********************************************************
skipping: [192.168.56.104]
skipping: [192.168.56.105]
ok: [192.168.56.108]

TASK [enterprise_service : Configure NGINX Port (Debian/Ubuntu)] ***************************************************
skipping: [192.168.56.108]
changed: [192.168.56.104]
changed: [192.168.56.105]

TASK [enterprise_service : Configure NGINX Port (RedHat/CentOS)] ***************************************************
skipping: [192.168.56.104]
skipping: [192.168.56.105]
ok: [192.168.56.108]

TASK [enterprise_service : Ensure NGINX is enabled and running] ****************************************************
changed: [192.168.56.108]
changed: [192.168.56.105]
changed: [192.168.56.104]
```

Playbook summary

```
PLAY RECAP *********************************************************************************************************
192.168.56.104             : ok=2    changed=0    unreachable=0    failed=1    skipped=1    rescued=0    ignored=0
192.168.56.105             : ok=2    changed=0    unreachable=0    failed=1    skipped=1    rescued=0    ignored=0
192.168.56.108             : ok=16   changed=2    unreachable=0    failed=1    skipped=9    rescued=0    ignored=0
```

5. For your final exam to be counted, please paste your repository link as an answer in this exam.
GitHub: https://github.com/mo-na-min/Final_Exam_Due-as
Note: Extra points if you will implement the said services via containerization.

2. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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